

Qatar-4b

R-2427

Benchmark run · workflow W8-benchmark-deep · record deep-bench_Qatar-4_221127 · created 2026-08-22T00:16:42+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2459910.7261 +/- 0.0035 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.156 +/- 0.018
Transit depth (Rp/Rs)^2	2.42 +/- 0.56 [%]
Orbital Inclination (inc)	85.46 +/- 1.68
Airmass coefficient 1 (a1)	1.008 +/- 0.02
Airmass coefficient 2 (a2)	-0.004 +/- 0.017
Scatter in the residuals of the lightcurve fit is	2.0 %
Best Comparison Star	#2 - [350, 443]
Optimal Aperture	3.56
Optimal Annulus	7.92
Transit Duration (day)	0.08 +/- 0.011

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.156 ± 0.018 vs 0.138 (deviation 0.98σ)
Duration fit vs published	0.08 d vs 0.0892 d (deviation 0.82σ)
Mid-time O–C	-3.4 min
Depth SNR	8.5
Residual scatter	2.0 %

Light curve

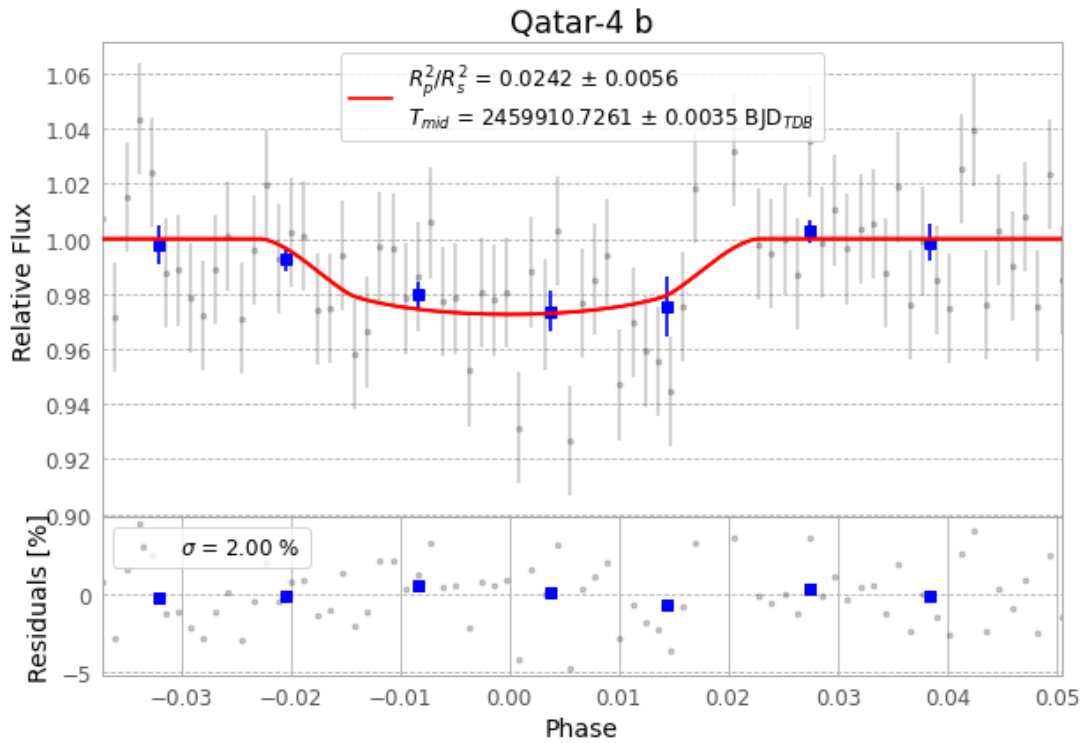


Figure 1 — FinalLightCurve_Qatar-4 b_2022-11-26.png (SHA-256 4741f0ba279dde0f...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [464, 227], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 b8cf9a5ea311a96e...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit a4253ce

Data provenance

77 FITS frames (51 MB), Qatar-4221127034215.FITS → Qatar-4221127073015.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-Qatar-4-221127.log (697700cba1b5...), FinalLightCurve_Qatar-4 b_2022-11-26.png (4741f0ba279d...), FinalLightCurve_Qatar-4 b_2022-11-26.csv (40484ebd639d...)

Compute resources

Wall time 113.0 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-22 00:16Z.